

Michael Mesarcik

Postgraduate Student

Contact

Cape Town

+27

Github
Website
Email

Languages

First Language:
English
Second Language:
Czech
Afrikaans fluency

Programming

Java
C
Python
C++
Android
VHDL
Verilog
MATLAB
Octave
Extensive unix usage

Architectures

STM32F0
STM32F4
Arduino
Embedded Computers
Altera Cyclone II
Altera MAX 10
Xilinx Artix 7

Education

- 2017–now **Masters of Engineering** The University of Cape Town, South Africa
Radar Engineering
With specialisation in Digital Signal Processing, FPGA Development and Radar
- 2013–2016 **Bachelor of Engineering** The University of Cape Town, South Africa
Computer Electrical Engineering
With specialisation in Digital Signal Processing, Digital Systems, and Mobile Broad-band Networks.

Experience

- 2017 – now **Analog Electronics Teaching Assistant** Cape Town, South Africa
Employee of The University of Cape Town
Assisted in the teaching of a second year analog electronics course. My responsibilities were to create and conduct tutorials and assignments as well as managing the administration of the course. In doing so, I gave lectures about using circuit simulation software as well managing a group of tutors.
Contact: Associate Professor Daniel O'Hagan
- 2015–2017 **Rosetta Roastery** Cape Town, South Africa
Developer
Developed a coffee roast monitoring system that tracks the development of beans being roasted. The project was executed using a Micro-controller, Adafruit Thermocouple and a multi-threaded cross platform PyQt based UI which interfaced using serial communications.
Contact: Jono Le Feuvre +27
- 2014–2015 **Thingking** Cape Town, South Africa
Summer Intern
Developed a digital periscope for a retail store in Johannesburg using a Unix backend for interfacing with an Maxbotix Ultrasonic Rangefinder.
Contact: Marc Nicolson +27
- 2014–2015 **Computer Science Tutor** Cape Town, South Africa
Employee of The University of Cape Town
Tutored both First and Second year Computer Science courses. The first year course was an introduction to programming concepts in Python, with tutorials varying from simple python constructs to recursion.
The Second year course consisted of modules in parallel programming in the Java Fork-Join Framework, computer architecture, and mobile application development using Android.
Contact: Kevin Elliott
- 2013–2014 **MacroLan** Cape Town, South Africa
Summer Intern
Observed and assisted Network Administrators and joined Network technicians on service calls, as well as helping in the technical support department.
Contact: Paul Johnson +27